

Production I: Firms and the Shock

Advanced Microeconomics 1: Economic Decision Making. Lecture 5

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Where we left the question

- L1–L4 answered the block's questions for households: who the 2022 shock hurt, by how much, and whether one-basket indexation is the right compensation.
- But we took the price change as given, $p^0 \rightarrow p^1$ just happened. And we treated indexation purely as a payment to households.
- Both are half the story. The same shock hit **firms**, as an input price, and indexation raises wages, which are (part of) a firm's costs.
- This block: the supply side.

The same shock, from the other side

- Energy is an input.
- BASF — Europe's largest chemicals maker, whose Ludwigshafen site is one of the biggest gas consumers on the continent — took an €800m hit from rising gas prices in a single quarter. Its CEO: “energy prices will stay at a high level and won't go back to normal soon.” Aluminium producers warned of “production cuts, job losses, and possibly a complete breakdown.”
- So: did European industry collapse in 2022?

It did not

- In 2022, EU industry cut natural-gas use by 15% against the 2019–21 average.
- Over the same period, industrial output rose by around 5% (Bruegel, 2023).
- Firms cut gas, not production. They substituted away from the input that got expensive.
- Exactly analogous to a household substituting away from the expensive good.

Today's route

1. Formalise the firm.
2. The profit maximization problem: $\pi(p)$ and $y(p)$, Hotelling's lemma.
3. The law of supply.

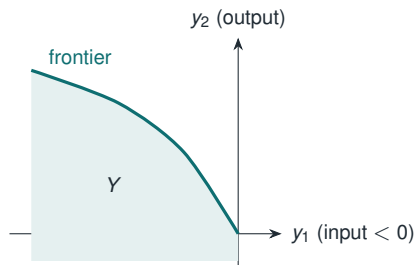
Formalising the firm

Production vector

Definition · production vector and production set

A **production vector** is $y \in \mathbb{R}^L$: $y_\ell > 0$ means output of good ℓ , $y_\ell < 0$ means input. The **production set** $Y \subseteq \mathbb{R}^L$ collects all technologically feasible y .

- A bakery:
 $y = (-\text{gas}, -\text{labour}, -\text{flour}, +\text{bread})$.
The 2022 shock is a price change on a negative component.
- Profit is $p \cdot y$.



Properties a production set can have (MWG 5.B)

Property	Content
Y nonempty, closed	limits of feasible plans are feasible
No free lunch	no outputs without inputs
Possibility of inaction	$0 \in Y$: can shut down
Free disposal	can absorb extra inputs harmlessly
Irreversibility	can't run production backwards
Returns to scale	nonincreasing / nondecreasing / constant
Additivity	$y + y' \in Y$: replication across plants
Convexity	mixtures of feasible plans are feasible

Describing the frontier

- **Single output:** $q = f(z)$, inputs $z \in \mathbb{R}_+^{L-1}$; $Y = \{(-z, q) : q \leq f(z)\}$.
- Slope of an isoquant — the **marginal rate of technical substitution**:

$$\text{MRTS}_{\ell k}(z) = \frac{\partial f(z)/\partial z_\ell}{\partial f(z)/\partial z_k}$$

how much more of input k you need when you give up a little of input ℓ , holding output fixed. The producer's MRS.

- Returns to scale for f : $f(\alpha z) \gtrless \alpha f(z)$.

The profit maximization problem

How does a firm choose its production plan

- Same as in L2: hypothesise the firm picks its **most profitable feasible plan**.
- For the consumer this was hard: utility is invisible, choices only reveal a ranking. For the firm it is “easy”: the objective is **observable and cardinal**: Profit.
- And there is **no budget constraint**. No wealth effects, no Giffen goods, no compensation subtleties.

The profit maximization problem

PMP

Given $p \gg 0$ and Y (nonempty, closed, free disposal):

$$\pi(p) = \max_{y \in Y} p \cdot y, \quad y(p) = \{y \in Y : p \cdot y = \pi(p)\}.$$

- π : **profit function**; $y(p)$: **supply correspondence**.

Proposition · MWG 5.C.1

For Y closed with free disposal:

1. π is **homogeneous of degree 1**; y is **h.d.0**.
2. π is **convex** in p .
3. **Hotelling's lemma**: if $y(\bar{p})$ is a singleton, $\nabla \pi(\bar{p}) = y(\bar{p})$.
4. If y is differentiable: $Dy(\bar{p}) = D^2 \pi(\bar{p})$ is **symmetric, positive semidefinite**, $Dy(\bar{p}) \bar{p} = 0$.

Proofs: MWG 5.C — each is the mirror of one from consumer theory.

The law of supply

Proposition · law of supply

For any p, p' and $y \in y(p), y' \in y(p')$:

$$(p - p') \cdot (y - y') \geq 0.$$

Proof. Optimality at p : $p \cdot y \geq p \cdot y'$. Optimality at p' : $p' \cdot y' \geq p' \cdot y$. Add and rearrange.

□

- Output rises when its price rises; **input demand falls when its price rises**. Gas gets more expensive $\Rightarrow$ use less gas.
- And **no compensation required** — contrast to the law of demand which held only for compensated changes. Why: no budget constraint $\Rightarrow$ no income effect to fight the substitution effect.

Answering (a): is the hypothesis testable, and does it pass?

- For a supply *function*: rationalisable $\iff Dy = D^2\pi$ symmetric PSD. The firm-side integrability — and no compensation needed, because there is nothing to compensate.
- For *finite data*: the firm's GARP is **WAPM** (Varian 1984): $p^t \cdot y^t \geq p^t \cdot y^s$ for all s, t — every observed plan beats every other observed plan at its own prices. Easier than the consumer's test in every way.
- The gas-cut-output-up pattern is exactly what the law of supply demands of an optimising input demand. The catastrophe headlines were a prediction of the theory.

Where the question stands

Takeaways

- **(a), first pass:** price-taking profit maximisation is testable *without compensation* — law of supply, symmetric PSD Dy , WAPM on finite data.
- **What we cannot yet do:** separate *economising on gas* from *scaling back output*, or say what indexed *wages* do to costs, prices and jobs. Both need the cost function.
- **Ahead:** Thursday, $c(w, q)$ — the EMP's twin .